ICT2113 SOFTWARE REQRS: NOW & NEAR FUTURE

10. Overcoming Deficiencies of RE – UserStory is Answer

Prof Kevin

“The moment you think you got it figured, you’re wrong.”

Shooter, 2007

Only one truly correct fact: User stories originated in XP SCM. But, it’s totally incorrect that super-simple format “As a… I want to … so that …” was ingrained at its creation. For XP developers, it was **ignorance gone viral**. To correct that misinformation, XP co-creator published effective writing convention decreeing that User stories are **social**, **casual**, and entirely **free-form**; no format is ever to be used. Incongruously, these two conventions of format & none, exist now harmoniously as **canon**. To exacerbate already paradoxical situation, one of early XP coaches then publishes 3rd canon that User stories are effectively written only when they conform to **quality control** features. Truth is out there!

Agenda: User story is answer to RE

1. Previous week's learning.
2. Today's learning outcomes.
3. Video – "User Story, Epic, Features, et al".
4. Today's new learning material.
5. Active learning – teams discuss response to ques.
6. Epilogue.

Please be ready to discuss with your team mates offline.

Even better, get together physically while using our zoom meeting.

Previous week's learning

Four slides

Remember why review of previous week's learning important!

Strengthen memory retention.

Highlight knowledge gaps.

Prepare for acquiring new material (per iterative learning approach in HE insti-tutions).

Guest lecture & RE (BRUF) review

Summary of Guest Lecturer's message. Notes taken by teacher.

Review of …

- ❑ … complete RE model and definitions of all four subprocesses, i.e., Eliciting, Analysing, Specifying, and Validating.
- ❑ … dominance of RE in sw development.

Summary of Guest Lecturer's message

Two activities are being recognized as key benchmarks in reqrs capturing.

- Testing of code by stakeholders. Compatibility to Gherkin format is key for success.
- Iterations, starting with 'minimum viable product' (MVP) and progressing through release functionality according to some prioritization schema.

Most of small and flexible apps, such as Spotify, are developed using Agile approach. Some are now getting customers & non-techies to write User stories for they improving functionality that they desire.

State of present reqrs capturing is as follows.

- Since they are very small, volume of User stories can be unwieldy to manage. Yet, they are still integral to Agile reqrs capturing; tools available for their management.
- Championed by Alistair Cockburn, Use case textual rendition (description) is emerging to prominence. In fact, it can be reqrs specification by itself.
- AI being applied to RE specifying and validating. Better, more in-depth training of AI such as ChatGPT is expected to bring great improvements to reqrs capturing process.

RE model: Methods, specifications & datastore

Note two cooperating orgs, four methods, two specs, and one datastore.

Customer's PL team	Vendor's Reqrs analysis team

Drawing forth lingering ideas & expectations from stakeholders on reqrs for new not yet existing sw, along with survey of pertinent historical record for presently-in-use sw, plus terms for new sw dev project, and then documenting all that info as clauses and models in Project Spec (PS).

Regulated-revision of PS into ready-content for SRS that is checked, fixed for integrity and comprehensiveness, plus enhanced with vendor's best practice in domain & technology. Relies on customer-vendor cooperation to clear up any uncertainties about reqrs.

Ana

Method

ation

re (VCS)

Writing sw reqrs spec (SRS) that comprises 'base' reqrs per agreed standards with codified 'development provisions' best practices, i.e., latest tech, sw dev quality model, iRTM & change mitigation strategy, all needed by vendor in post-reqrs SDLC.

Release to contracted vendor

Conducted in-house or by ext consultant

Applying quality control measures onto SRS, and conducting quality assurance audit of vendor's conduct of Analysing, Specifying, and Validating against contractual and professional standards, to ensure that vendor has right SRS produced right.

(UML Activity diagram)

Dominance of RE in sw development

In mid-1940s, first generation of digital electronic computers built by Electrical and Electronic engineers according to hardware development standards, because software as we know it now didn't exist then. Computing cycles were run according to physical rearrangement of patch cables and manual switches.

1950s saw emergence of sw being an internally stored program that enabled computing cycles to be run without manual setup. This new sw was written by new breed of gifted pseudo-professionals who approached each new project on ad-hoc basis, neither baselining, repeating, nor optimizing their code.

By mid-1960s, sw industry was in crisis situation of massive project delays and abandonments. In 1968, much needed discipline was established via SWE and SDLC. Unfortunately, engineering rigour and quality that was being applied to design, coding, testing, and deployment, came late to capturing requirements; 1997 in fact. By then, RE was established as 'the actionable methodology' that aligned with any prevailing 'software development methodology' (SDM) that called for 'big reqrs up front' (BRUF).

BRUF is associated with waterfall model and other inceptive models of SDLC.

Today's learning outcomes

One slide

Remember why knowing learning outcomes important!

Roadmap of learning goals & expectations.

Encourages develop-ing preparation & study strategies.

Promote engaging in learning process.

Facilitate managing learning material & effort.

LO#10: User story is answer to RE

After today's lesson, student will be able to explain …

- … quiet revolution of RCM from heavyweight to lightweight pre-Agile conclave.
- … User stories method & methodology.
- … origin of User stories.
- … importance of Agile conclave on dominance of User stories.
- … three conventions for effectively writing User stories.

After today's lesson, student will be able to sketch artistically as opposed to formally User story process model.

These are objectives I plan for you to achieve in this session.

Your objectives should sync with these, but don't need to be verbatim identical.

Video

One slide & 7:09 min clip

Clip10 User Story, Epic, Feature, et al

Things to keep in mind before viewing

Video is brief explanation of User stories, plus most of size categorization terms associated with .

- First video defines differences between software development methodologies that are traditional waterfall-compliant vs those being Agile.
- Second video is about present ad hoc approach to organizing User stories.

As you watch first video, please keep foremost in your mind that software development drives innovation in requirements capturing. Up to now, this is consistent historical record.

For second video, ask yourself why it had to made. User stories are innovative, but do they more surpass RE? No; seems, there is weakness with User stories that our industry is responding to. What is it? Do you agree with “tourniquet” solution.

Signal when you understand, and are ready to begin.

Today's new learning material

Eighteen slides

User stories: Revolution of sw proj weighting

To understand significance of User story, we must appreciate weighting evolution from heavy to light that SDM-RCM pairing underwent. Since inception of modern sw in 1950, SDM was heavyweight and suitably paired with RE RCM. So it remained pre-eminent for several decades of sw projects.

In 1970s, lightweight SDMs started emerging. Being paired with SDMs, RCM began changing also, though not quite as quickly or substantially as SDMs. Strict linear arrangement of RE gave way to incremental, iterative, and even evolutionary. Feedback between customer and vendor increased; some methodologies even captured reqrs jointly. Though innovative, these RCMs still retained vestiges of traditional practice.

Finally, 1996 saw first truly lightweight RCM (see next slide). To save floundering Chrysler Comprehensive Compensation (C3) project, PM Kent Beck created never seen before SDM & RCM. Though largely unnoticed at first, within three years it would become mainstay of lightweight RCM.

(See next slide) SDM evolution traces back to 1956 & is still going strong. What came before 1996 was great, but XP surpassed them all.

User story is wonderkid of RCMs.

Key pioneering SDM innovations (w/RCM)

Code
D = Sw develop
M = Proj Mgmt

(M, 1957) Incremental, by Weinberg & Dimsdale in IBM

(M, 1977) Joint Application Dev, by Chuck Morris & Tony Crawford in IBM

(M, 1977) V-chart of SDLC, by J.B, Munson in Sys Dev Corp

(D, 1986) Spiral Model of SDLC, by Barry Boehm in TRW

(M, 1956) Kanban, by Taiichi Ōno in Toyota

(M, 1991) Rapid Application (RAD), by James Martin in IBM

(M, 2007) Knbn Method, by Anderson in Corbis

(D, 1997) Feature-Driven Dev, by Jeff DeLuca in United Overseas Bank

(D, 1996) **Extreme Programming (XP)**, by Kent Beck in Chrysler

(D, 1986, 1995) **Scrum**, by Schwaber & Sutherland

(M, 1994) Dynamic Sys Dev Method (DSDM), by Consortium in pilot projects

Innovation gallery

- Min viable product, then fcn after fcn
- Continuous participation of customer
- Continuous multi-level feedback
- Build in rapid cycles
- Work tracked at regular intervals for JIT
- Fast delivery of working program
- Visibility of progress & results
- Best practice to extreme
- Increased flexibility & speed
- Continuous & clear comms

Lightweight RCM employing **collection** of **User stories**, or short, self-contained 'stories of usage by stakeholders', that are "markers" for full reqrs of new not yet existing sw. It comprises **three** steps, 1) **write** User stories, 2) **bring forth** via **conversation**, **full specification** of reqr behind User story's brevity, and 3) **prioritize** User stories for **selection from PB** and **realization**. Once realized, User stories are discarded, and new **source code** becomes primary documentation for new sw. All three methods in this RCM involve User Story artifact only, justifying designating this "the User story methodology".

User Story methodology

Second of three Reqrs Capture Methodologies that we will study our module.

User story: Anchor element of this RCM

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Informal description of new sw's capability/condition from perspectives of its user, imagined usage, and expected benefit, along with its production-grade code being realized (designed, coded & accepted by stakeholder) within hours to few days.

Factsheet of fundamental ideas about User story RCM

Conversation. User story is enough to initiate expansive conversation (verbal, docs, observing) between stakeholder & developer to specify reqr. Little saved after realization: source code & some design.

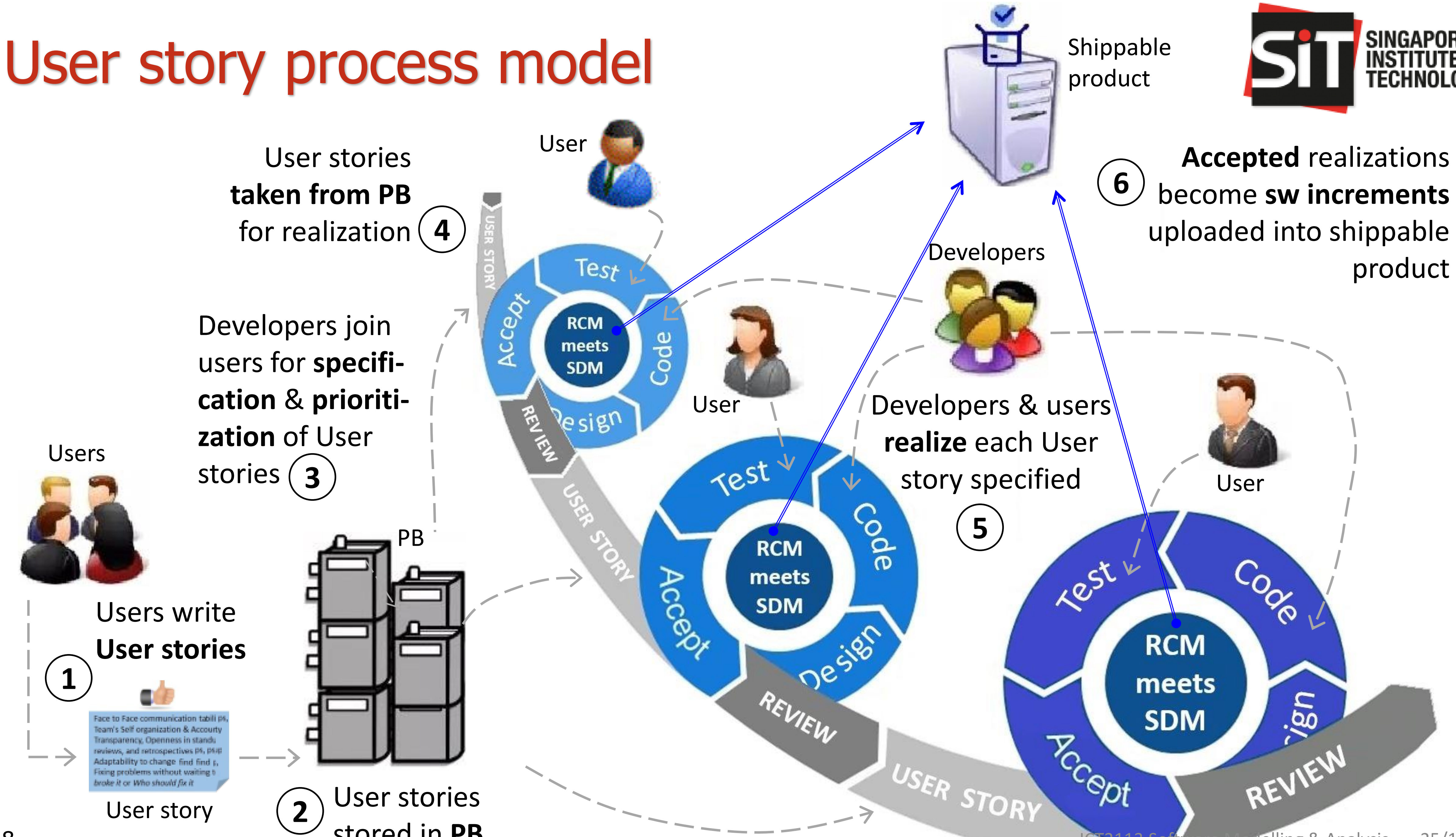
Epic & Feature. Each sw dev proj has many user stories, 100s even 1,000s, because each applies to very small part of new sw. To cope with such, often user stories amalgamated into "Epics" to better manage numerous disparate tasks, or "Features" to optimally integrate code blocks into functions. These are broken apart to be realized as individual stories, only to be rejoined post realization.

Product backlog. Stories held in repository 'Product Backlog' (PB) to await realization; no SRS per se.

Not saved. Once chosen for realization, User story deleted from PB.

Coach. Typically, first-time customers hire 'Agile coaches' to cope with gap between Agile and corporate norms. Biggest tension is with fragmentation debt: system context lost in sea of disjointed PB items.

User story process model



User story's origin is embedded in XP practice

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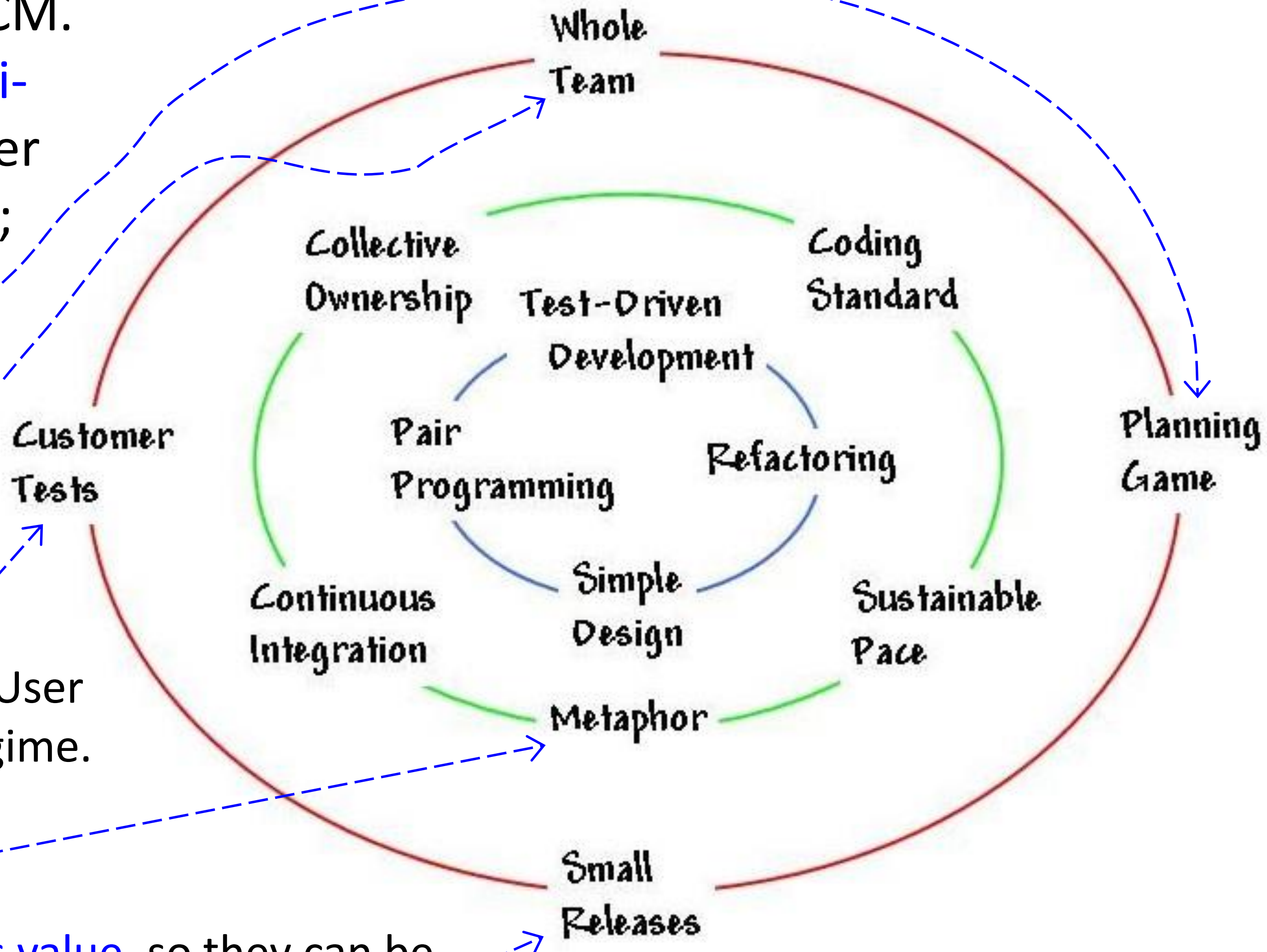
User stories are super-tightly bound to XP SCM. Of original 13 practices, five (see right) are directly driving User stories. For XP teams, User stories are written as they always have been; effectiveness is non-sequitur.

- Customer conducts release planning, i.e., difficulty, cost & scheduling of realizing User stories.
- Real users are part of whole team for reqrs, priorities, and steering of proj.
- Customer defines acceptance tests for realized User stories; theses added into regression testing regime.
- Proj co-opts Customer's simple evocative vision of how new sw works, via use of metaphor.
- Customer sets reqrs small yet with real business value, so they can be realized and shipped to customer frequently as "shrink-wrapped" sw features.



Anything you are uncertain about? …

Ask, and let me help clarify

Take quick break

300 sec countdown – ends when fully filled

Let's continue with LS

User story effective writing: 3xConventions

Though User story is rightfully considered "wonderkid of RCMs", it has one significant issue: how to effectively write it. XP developers treat story writing as "leave well enough alone". Developers from other SCMs would be unacquainted with stories, ergo need writing guidance. Presently, three conventions recognized (ignoring many one-off variations arising since).

1. Original no name → Three-C's – forged from crucible of real practice in XP.
2. Connextra (via format) – brought in as coaching aid.
3. INVEST (via quality) – benchmarked against tailored effectiveness features.

1. Original no name

After its launch in 1993, C3 project SDM suffered setbacks. Within one year of being new PM (Mar 1996), Beck brought C3 back on track to pay 10K employees. Key to his success was replacing BAs with business users to articulate C3's real needs, and letting them do it in story form directly to developers. Written on 3"x5" index cards, stories were bereft of detail deliberately to engage developers in superlative conversation with users.

What happened to C3 project? It was halted early 1999 pending Daimler-Benz's buyout of Chrysler.

New owners brought in their own compensation sw and summarily terminated C3 in Feb 2000.

Even so, XP is seen as triumph.

1. Original no name (cont)

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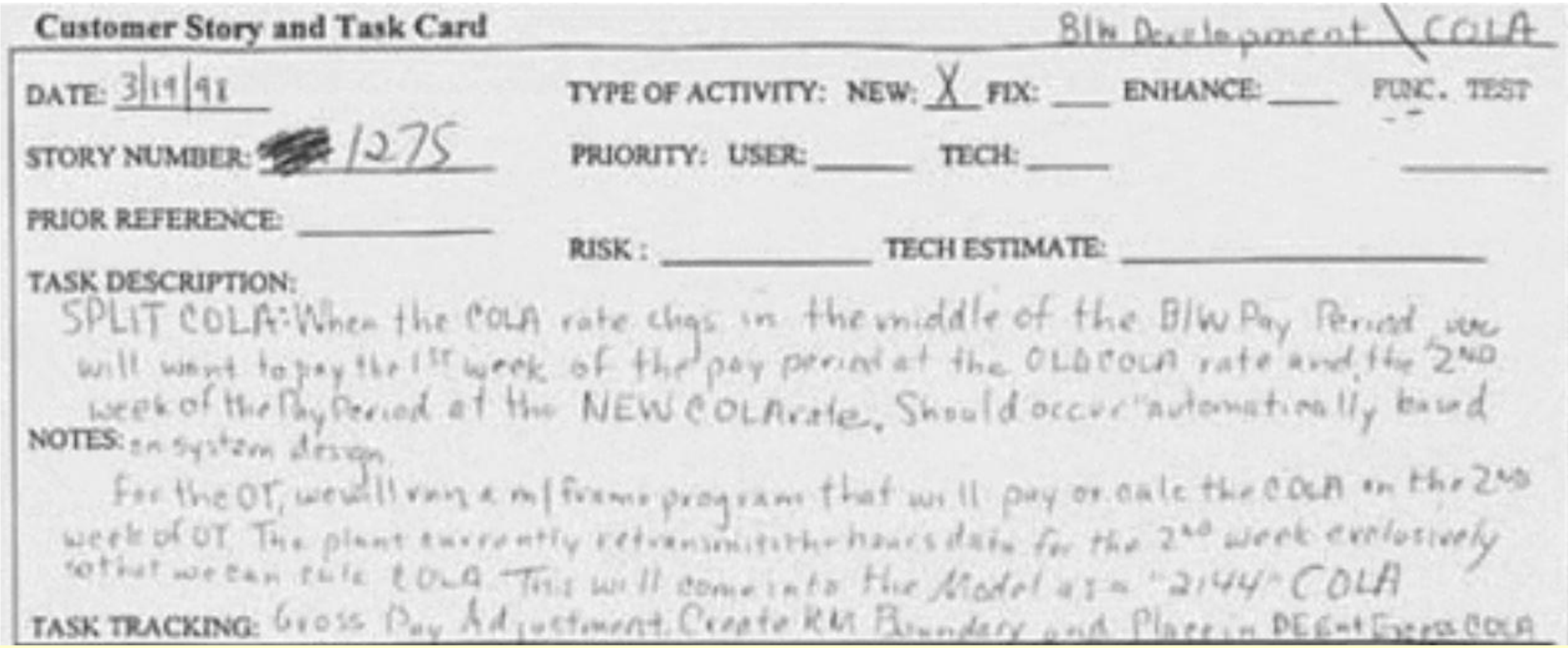
Crucially, said stories were unformatted & force-fit on index cards (see right; from Beck's 1999 book) so users don't infer they are writing full-blown reqrs for SRS. Size of cards used was never more than 3"x5", but could be smaller to force large stories to reduce.

User story always intended to initiate conversation and collaboration.


Customer Story and Task Card — B/W Development \ COLA

DATE: 3/19/98 — TYPE OF ACTIVITY: NEW: X FIX: ___ ENHANCE: ___ FUNC. TEST

STORY NUMBER: 1275 — PRIORITY: USER: ___ TECH: ___

PRIOR REFERENCE: ___ — RISK: ___ TECH ESTIMATE: ___

TASK DESCRIPTION:

NOTES:

TASK TRACKING:

SPLIT COLA: When the COLA rate chgs in the middle of the B/W Pay Period, we will want to pay the 1st week of the pay period at the OLD COLA rate and the 2nd week of the Pay Period at the NEW COLA rate. Should occur automatically based on system design.

For the OT, we will run a m/frame program that will pay or calculate the COLA on the 2nd week of the OT. The plant currently retransmits the hours data for the 2nd week exclusively so that we can calc COLA. This will come into the Model as a "2144" COLA.

Gross Pay Adjustment. Create RM Boundary and Place in DE Ent Express COLA BIN.

1. Original no name (cont)

Alistair Cockburn, independent consultant & researcher focusing on successful SDM, visited C3 project in 1998, expecting Use cases. He wrote "They put so little into each use case that they called them "stories" rather than 'Use cases', and wrote them on single cards. Each story card was really a promise for a conversation between a requirements expert and a developer." (Cockburn, *Writing Effective Use Cases*, 2000/2001, p. 223) It highlights value principle of 'human interaction over documentation' that later became part of 2001 Agile Manifesto.

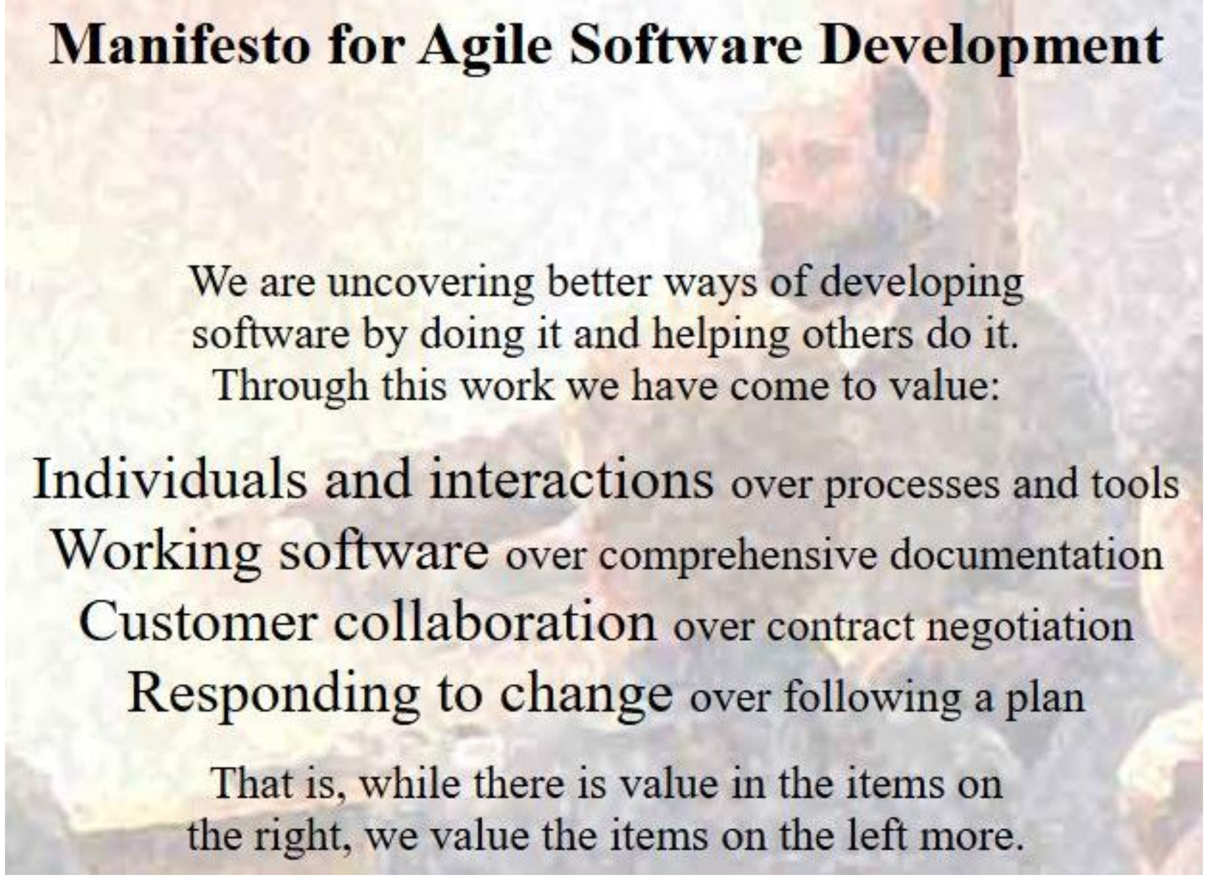
Three credited founders of XP SDM are Kent Beck, Ward Cunningham & Ron Jeffries.

Then followed flurry of publications on XP SDM and transformative nature of its RCM (user story).

- In 1999 Beck, as creator of XP SDM, authored "XP Explained: Embracing Change".
- In collaboration with Martin Fowler, prominent thought leader in OOP underpinning core of XP practice, Beck publishes in Oct 2000 his second book "Planning XP".
- After C3 terminated, Ron Jeffries turned to independent consultant to promote XP & coach early adopters. In Nov 2000, he published "XP Installed" with two XP coaches Ann Anderson and Chet Hendrickson.
- In Jan 2001, independent consultant & XP coach Bill Wake published "XP Programming Explored" for applying XP best practices.
- Aug 2001 Ron Jeffries posts seminal blog on User story effective writing practiced in XP projects.

Agile arrives: "Vive la révolution"

On 11–13 Feb 2001, seventeen pioneering innovators met at Snowbird resort in Utah to discuss lightweight SDM: *Kent Beck*, *Ward Cunningham* & *Ron Jeffries* (XP), *Jeff Sutherland*, *Ken Schwaber* & *Mike Beedle* (Scrum), *Dave Thomas* & *Andrew Hunt* (Pragmatic Programming & Ruby) , *Jim Highsmith* (ASD), *Alistair Cockburn* (Crystal Method), *Robert C. Martin* (Object Mentor), *Martin Fowler* (OOP, UML & XP), *Arie van Bennekum* (Lightweight consult), *James Grenning* (XP, embedded & TDD), *Jon Kern* (Adaptavist Grp), *Brian Marick* (Ruby & TDD), and *Steve Mellor* (OMG).



After conclave ended, self-named 'Agile Alliance' published manifesto (see above right) for their 'agile approach' — comprising core values and principles — to sw development. 'Agile' now synonymous with 'Lightweight'; terms are interchangeable. Immediately, list of SDMs that fully adhere to Agile manifesto was agreed: ASD, Crystal Clear, DSDM, FDD, JAD, Kanban Method, RAD, Scrum & XP. Same was done for RCMs, although list had only one entry: User story from XP. So began User story's reign as 'Agile RCM'.

Agile arrives: User story fits SDM revolution

Five reasons why User story is outstanding fit for being Agile RCM.

1. Its focus is 'human interaction over documentation' which fits almost word-for-word with principle of Agile Manifesto.
2. It pairs with all Agile SDMs. This is because User story captures reqrs iteratively, adaptively, flexibly & via continual feedback with stakeholders, all synonymous with lightweight. Plus, User stories carry nil technical detail, making them totally opposite to BRUF hallmark of heavyweight SDMs.
3. It can be written by any stakeholder type, especially by direct & indirect end users that often were ignored in heavyweight RCMs.
4. It transcends language & skill constraints. This is because User story captures story-like descriptions of whatever stakeholders sees and does in natural and non-technical language.
5. Testing for acceptance of newly-crafted sw is conducted by same stakeholder that composes User story, together with same developer that produces its production-grade code; another hallmark of lightweight.

2. Connextra format: coaching aid

After Agile conclave in Feb 2001, demand for Scrum and XP SDMs grew, but not without growing pains. Aside from lack of qualified Scrum/XP coaches, existing problem of effectively writing User stories worsened as many more "new-to-Agile stakeholders stuck in BRUF approach" needed Agile coaching.

In May 2001 during their new sw installation project for UK company Connextra, Rachel Davies and her XP development team conceived and applied new format template for effectively writing User stories. Named "Connextra format", it is sentence comprising three parts that are filled in by stakeholder (see below; underlined & italicized words indicate type of entries by stakeholders).

> As a *position* , I want to *do biz process* , so that *benefit derived* .

Perfectly timed (few months after conclave), it was quickly adopted as practice standard for effectively writing User stories. Later, it was formalized in Mike Cohn's 2004 "User Stories Applied for Agile Software Development.

Cohn's 2004 book is considered definitive treatise on User stories, and earned him placement alongside Agile luminaries.

Interestingly, no one in Davies' team ever wrote about conceiving & applying Connextra format. So, when Cohn credited Davies' team in his book, he became sole source of that historical account.

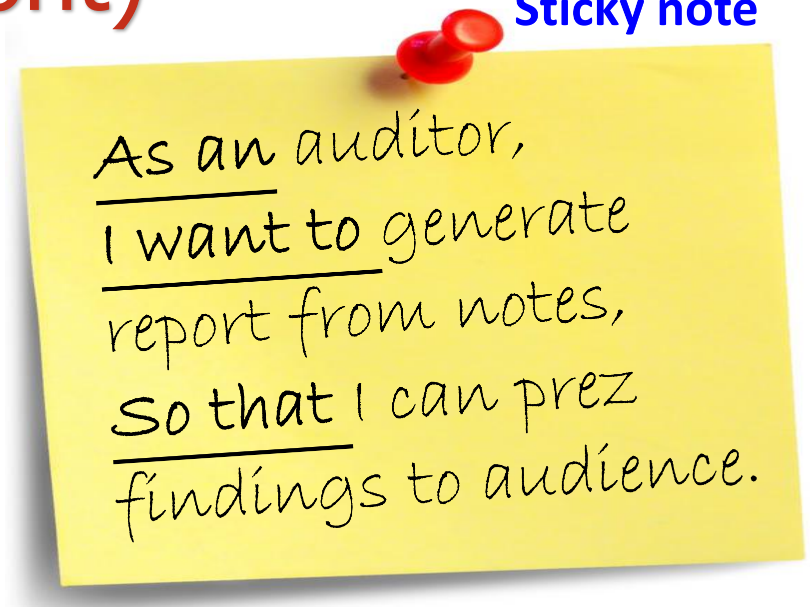
2. Connextra format: coaching aid (cont)

(See right) First, format written on 'sticky note' & '3"x5" index cards' for physical placement on 'progress board'. Later, transitioned to data elements on digitized flow charts & maps. Tips for content style in each entry field :

- **Position**. Must be specific job appointment or title. Generic roles such as "user" or "manager" are insufficient to under-stand context and identify stakeholder (for conversation).
- **Biz process**. Work unit per breakdown of stakeholder's job duties; not technical action such as pressing button. It must have measurable criteria of completion, although that is not shown with story itself.
- **Benefit**. Rationale or justification that is job-related, not personal. It can be any of: achieving biz goal, fulfilling biz purpose, or generating something of value to biz.

Acceptance tests may be added to note or data element.

Sticky note



Data element

As a <bank> customer ,
I want to view my current account balance ,
so that I know my recent deposit went through .

2. Connextra format: coaching aid (cont)

These example user stories illustrate well-articulated biz processes and derived benefits for different stakeholders in different scenarios.

1. Scenario example 1: Product development.

   As a product manager, **I want** a way for team members to understand how individual tasks contribute to larger business goals, **so that** their commitment and cooperation are enhanced.

2. Scenario example 2: Customer experience.

   As a returning customer, **I want** my information to be saved, **so that** I have a more streamlined checkout experience.

3. Scenario example 3: Mobile application.

   As a frequent app user, **I want** a way to digest relevant information in the quickest way possible, **so that** I can decide on the best course of action in the fastest time.

User stories such as these are posted into PB, periodically prioritized, possibly aggregated into Epics or Features, and eventually realized-then-integrated individually into host product or program.

1. Original becomes "Three-C's"

Four books 1999 to 2001 (see slide 18) about XP never specified any convention for writing User stories. This was corrected three months after Connextra format became public, when Jeffries posted blog that reaffirmed XP's original non-format convention, and finally gave it name, "Card, Conversation, Confirmation" or "Three-C's". Also, he highlighted higher effectiveness of social reqrs capture over "documentary". 18 years later, Jeffries revisited first blog with two more in quick succession. As such, their language being casual, not precise, open to interpretation; mine is things derived in stages.

- **Card**. This is first-hand outline of biz process with just enough info to evoke one unique proto-reqr to be specified in full in subsequent stage (Conversation). It is bound not by standard format, but by standard-size text box. If it exceeds standard size, then it must be revised to fit into smaller box. Its author must be authentic user who stays for all of three-C's. Media that stores outline can be physical or digital, as dictated by PB setup. Finally, if validated during Conversation, User story in its standard-sized box, is used as marker in mapping its progress through realization. (Mapping is conducted on Story board.)

Jeffries refers to Card, Conversation, Confirmation as "components". So, considering them as things seems right.

Also, Jeffries refers to select entities in one component being planned in previous component, suggesting sequence of stages. Yet, Card is optional, so staging may be oversimplification.

1. Original becomes "Three-C's" (cont)

- **Conversation**. This is collaboration between user and developer akin to Eliciting Method in RE RCM. Its purpose is to specify full reqr from 'proto' one and to estimate realization effort. It uses many "social" eliciting techniques, exchanges documents & examples, and demos lots of prototypes. Before doing specification, User story's proto-reqr is validated — it can be outright accepted, or discarded entirely, or replaced by revised story. Notes concerning priority and cost can be appended to validated User story, depending on available space. As specification is being conceptualized, user is also generating all criteria that will used to accept sw realization of reqr specification tied to User story. Collaborating on said criteria helps to bring specifications closer to perfection, avoid reqrs creep, and optimally shorten overall realization duration.

- **Confirmation**. This is realization of sw by developer, followed by acceptance testing of said sw by authentic user. This akin to SDLC User Acceptance Testing (UAT) with two important differences: a) user already familiar with sw's capability and UX because pre-coding development work is done in full collaboration with developer, and b) only one reqr is being tested because stories realized one-by-one, followed by release of sw incements to customer. (This is way of lightweight SDMs.)

Achieving user's acceptance marks end of Three-C's convention. Next, sw enters process of integration into host product/program, involving different customer-vendor teams.

1. Original becomes "Three-C's" (cont)

- **Conversation**. This is collaboration between user and developer akin to Eliciting Method in RE RCM, to specify full reqr from 'proto' one. To do this, it uses any eliciting techniques, including interviews, exchanges of documents & examples, and prototypes:. a) validate proto-reqr story — yes, it can be outright accepted, or discarded entirely, or replaced by revised story — and b) then estimate realization effort — yes, notes concerning priority and cost can be appended to validated User story, depending on available space. that supports collaborations between user and developer. (In fact, User story is one recognized eliciting technique.) Said collaboration. Finally, this must produce criteria that user will follow for accepting sw realization of reqr specification of User story; importantly focus & limit to avoid reqrs creep.
- **Confirmation**. This is acceptance test of sw that developer realized for User story's reqrs specification produced in Conversation. This akin to SDLC Testing stage as acceptance is primary responsibility of customer. Difference is that a) user already familiar with sw's capability and UX because pre-coding development work is done collaboratively with developer, plus b) only one reqr is being tested.

Achieving user's acceptance marks end of Three-C's convention. Next, sw enters process of integration into host product/program, involving different customer-vendor teams.

3. Benchmark effectiveness via quality stds

Independent consultant & XP early practitioner Bill Wake published third convention of User story effective writing in Aug 2003 blog, “INVEST in Good Stories & SMART Tasks”. Acronym INVEST highlights characteristics that User stories of best quality will exhibit.

- “I” **ndependent**. Its content does not overlap with any others.
- “N” **egotiable**. Its exact representation of full reqr is completely open to being elaborated via conversation. (Assume “conversation” meaning as per Three-C’s.)
- “V” **aluable**. Its derived benefit to authentic user must be tangible & compelling.
- “E” **stimatable**. It must have enough prioritization content to enable cost & effort derivation by orgs.
- “S” **mall**. It fits in given standard text box and its realization is achievable in minimum period of time.
- “T” **estable**. Its operability must be measurable, so definitive acceptance can be made of its suitability.

Most projects that incorporate INVEST convention in their Agile RCM, stipulate that authors must accompany their User stories with scorecard that shows which quality features are met and which aren’t. Poor scorecard doesn’t mean User story is discarded; more often, developers use it to critique when and in what order such User stories are processed.

Which convention best for Agile RCM?

In 2004, Mike Cohn published book "User Stories Applied For Agile Software Development" that is considered by many to be canon of how to apply User stories in Agile SDMs. In it, he discusses how and when to apply all three effective writing conventions we have learned thus far.

He suggests following usage scheme (gross oversimplification of 291 pages):

1. Connextra format for customer's first few Agile RCMs. It is easiest to teach and vet. Most common error authors will make is being too generic, e.g., As a <u>user</u>, I want to <u>check filled-out form for mistakes</u>, so that <u>I avoid mistake propagation</u>.
2. INVEST for customers who no longer need special coaching for their RCM. It is more effective for increasing quality of User stories that are unformatted. Still, that doesn't preclude applying it to User stories that are formatted.
3. Three-C's for customers whose company culture has been re-oriented to proficiently conduct Agile SDMs, especially XP's. To be clear, Connextra format is not compatible with Three-C's. Also, developers have to be super-comfortable with talking reqrs with users.

Throughout mid-1990s, Mike Cohn was running rapid sw dev: no documents, no BRUF, just verbally engaging users, hand-drawing designs on napkins, and prototyping (lots). He always felt guilty that his way was not how it was supposed to be. Then, he read Kent Beck's 1999 book, and suddenly, all of his guilt evaporated.

Active learning 1-2

Two slides

These are instructions for session

There are two parts to today's 'Active learning' session:

Part 1 – Explain why in Three-C's if User story exceeds standard size, then it must be revised to fit into smaller size?

Part 2 – Propose your team's choice of User story effective writing convention and how that convention will be conducted and managed?

For Part 1. Clue is in factsheet and named convention's stipulations. Teams must explain exactly how size is measured, and what difference is between standard vs smaller.

Teams have five mins to come up with their answers. Once time is up, teams be prepared to share their results if called by teacher. Feedback is available from teacher and peers.

Any questions? Wait for teacher's signal to begin Part 1.

These are instructions for session (cont)

For Part 2. Teams can ignore Cohn's usage scheme, i.e., Connextra for beginners, INVEST for intermediates, and three C-s for experts. Choose whichever convention you want to try.

Crucially, don't combine conventions into some hybrid; adhere to their stipulations in their purest sense. Be especially precise for INVEST: if you decide formatted, it cannot be Connextra; you have to come up with your own format and stick to it. Also, don't write any User stories.

Teams have 15 mins to come up with their choice. Once time is up, teams be prepared to share their results if called by teacher. Feedback is available from teacher and peers. After LS, bring choice to this week's labtorial to begin applying it to your team's PD. Post notes.

Any questions? Wait for teacher's signal to begin Part 1.

Epilogue

Three slides

There is so much misunderstanding about User stories, because they emerged not from centralized revolution, but from counter-culture of innovation by engineers of many perspectives, who understood that developing sw was exceedingly different to that of hardware. User stories are paradigm shift in reqrs capture. They aren't reqrs, and they don't replace documentation. Instead, they represent ideas of reqrs, and ideas are fluid. So, User stories walk fine line of being written ideas that evoke tangibilities that are commonly comprehended, without being too verbose that they supercede conversation and collaboration.

"Just enough to represent, yet not stifle flexibility"

My final words

- Contact me for …
  - regular stuff via my work email h51581@singaporetech.edu.sg.
  - more sensitive stuff, by Whatsapp **97761716** or my personal email johnmilton2000@gmail.com.
- Tomorrow's lab is on this subject of Validating. This is the last part of capturing requirements via traditional RE methodology. In preparation, ensure you are familiar with …
  - your role, either QC or QA, including their relevant checklists and tasks.
  - other team's SRS that you will be conducting the QC and QA upon.
  - your plan (possibly posted in LMS) for catching QC and QA errors.
- You should be nearly through Glossary; good, keep going. Also, keep up with references to get deeper sense of modules objectives.

Glossary & Google reading for next week

- UseCase2.0
- OO SWE — A Use-Case-Driven Approach by Ivar Jacobson
- Writing Effective Use Cases by Alistair Cockburn

Any final thoughts?

If not …

End of LS#10
Overcoming RE Deficiencies – UserStory is Answer
by Prof Kevin

ICT2113 Software Reqrs: Now & Near Future